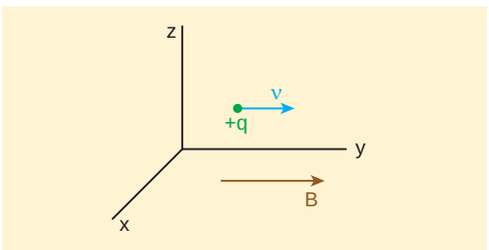
## EXAMPLE 3.20

A particle of charge q moves with velocity $\vec{v}$ along positive y - direction in a magnetic field $\vec{B}$. Compute the Lorentz force experienced by the particle (a) when magnetic field is along positive y-direction (b) when magnetic field points in positive z - direction (c) when magnetic field is in zy - plane and making an angle θ with velocity of the particle. Mark the direction of magnetic force in each case.

### Solution

Velocity of the particle is $\vec{v} = v\,\hat{j}$

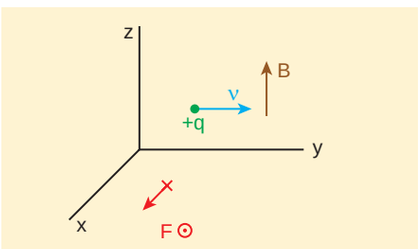
(a) Magnetic field is along positive $y$ - direction, this implies, $\vec{B} = v\,\hat{j}$




From Lorentz force, $\vec{F}_m = q\left(v\,\hat{j} \times B\,\hat{j}\right) = 0$

So, no force acts on the particle when it moves along the direction of magnetic field.

(b) Magnetic field points in positive z - direction, this implies, $\vec{B} = B\,\hat{k}$

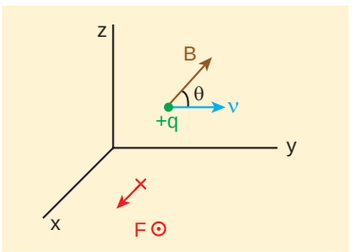



From Lorentz force,

$\vec{F}_m = q\left(v\hat{j} \times B\hat{k}\right) = qvB\,\hat{i}$

Therefore, the magnitude of the Lorentz force is $qvB$ and direction is along positive x - direction.

(c) Magnetic field is in zy - plane and making an angle θ with the velocity of the particle, which implies $\vec{B} = B\cos\theta\,\hat{j} + B\sin\theta\,\hat{k}$




From Lorentz force,

$\vec{F}_m = q\left(v\hat{j}\right) \times (B\cos\theta\hat{j} + B\sin\theta\hat{k})$

$= \text{qvB}\sin\theta\hat{i}$

## EXAMPLE 3.21

Compute the work done and power delivered by the Lorentz force on the particle of charge q moving with velocity $\vec{v}$. Calculate the angle between Lorentz force and velocity of the charged particle and also interpret the result.

### Solution

For a charged particle moving on a magnetic field, $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$

The work done by the magnetic field is

$$W = \int \vec{F}.d\vec{r} = \int \vec{F}.\vec{v}dt$$

$$W = q\int \left(\vec{v} \times \vec{B}\right).\vec{v}\,dt = 0$$

Since $\vec{v} \times \vec{B}$ is perpendicular to $\vec{v}$ and hence $\left(\vec{v} \times \vec{B}\right) \cdot \vec{v} = \vec{0}$ This means that Lorentz force do no work on the particle. From work-kinetic energy theorem, (Refer section 4.2.6, XI th standard Volume I)

$$\frac{dW}{dt} = P = 0$$



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